

Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 10
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

Title: Study of Communication Toolbox

Objectives:

1. To understand features of communication toolbox
2. To create waveforms using wireless waveform generator app
3. To add impairments in generated waveforms and analyze the effect of it.

OUTCOME:

1. understand working of analog communication transmitter and receiver systems
2. understand noise effect on the performance of communication systems.

1) Communication toolbox: <https://in.mathworks.com/help/comm/get-started-with-communications-toolbox.html>

- Enlist various features provided by communication toolbox

2) Wireless waveform generator app

- From the MATLAB App section, open wireless waveform generator app
- List down all the waveforms available in this app
- Select sine wave and visualize the waveform in time and frequency domain
- Add different impairments to this signal, take a output of each impairments and interpret the results. Take screenshots of all settings done
- Select any one other waveform from the list and repeat the same procedure for this waveform.

- **Conclusion:**

- **Post lab activity:**

- Why spectrum analysis important in communication system? What information we get from it? (2 marks)
- Enlist different impairments setting available in MATLAB Wireless Waveform Generator App and write your observations of adding this impairments to the generated signal (3 marks)

Signature of faculty in-charge

Department of Electronics and Telecommunication Engineering (Programme: EXTC)